

Activity No. 4

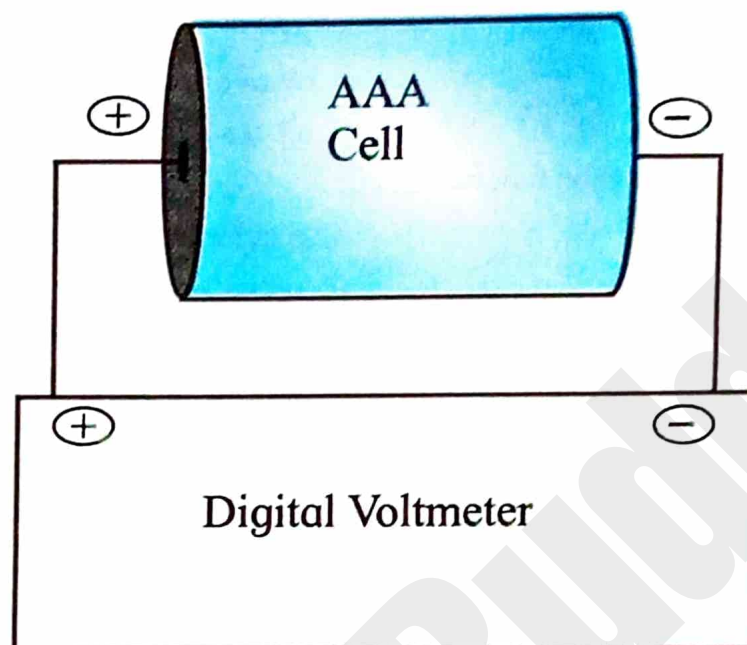
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Aim: Measurement of e.m.f. of commercial cells by using voltmeter.

Theory: There are different types or examples of Galvanic or Voltaic cells. In all these cells electrical energy is produced by chemical change, when these are connected in circuit.

Apparatus : AAA cell, AA cell, cell of wrist watch, any cell phone battery, voltmeter, wire, etc.

Diagram:



Procedure:

1. Connect the both ends of voltmeter by copper wire to positive and negative terminal of each given cell.
2. Measure the voltage shown by voltmeter and record it
3. Repeat the above procedure for all different cells provided.

Observation and Result:

Sr. No.	Name of the cell	Voltage observed in volt
1	T.V. Remote cell	1.6 V
2	Wall clock cell	1.6 V
3	Torch battery cell	3.7 V
4	Wrist watch cell	3.1 V

Remark and sign of teacher:

MCQ

Select [✓] the most appropriate answer from given alternatives of each sub question.

- Which one of the following device is used to measure e.m.f. of different types of cells?
☒ a. voltmeter b. barometer c. thermometer d. galvanometer
- The full form of abbreviation e.m.f. is.....
 a. electromagnetic field ☒ b. electromotive force
 c. electric and magnetic field d. electron messon field
- Which one of the following device will give most accurate reading of potential difference that is voltage?
 a. poggendroff's analog method b. voltmeter
☒ c. potentiometer d. multimeter
- The electromotive force is also called as
☒ a. e.m.f. b. cell potential
 c. standard electrode potential d. electro chemical series
- If e.m.f. of the cell is positive, the reaction is.....
☒ a. spontaneous b. non spontaneous
 c. endothermic d. exothermic

Short answer questions

1. Why multimeter or voltmeter does not gives more accurate reading of voltage?

Ans. It is because Voltmeter does not draw current from internal circuit and draws current from only two terminals of a cell

2. Write the name of the cell which is generally used in transistors.

Ans. The dry cell which is known as leclanche cell is used in transistor.

3. Write formula for calculating e.m.f. of the cell.

Ans. The formula for calculating e.m.f. of the cell is : $E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$.

4. Calculate the cell potential (Given: $E^{\circ}_{\text{cathode}} = 0.4 \text{ V}$ and $E^{\circ}_{\text{anode}} = -0.83 \text{ V}$).

Ans.

$$\begin{aligned}
 E^{\circ}_{\text{cell}} &= E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} \\
 &= 0.4 - (-0.83) \\
 &= 0.4 + 0.83 \\
 &= 1.23 \text{ V}
 \end{aligned}$$

Remark and sign of teacher: